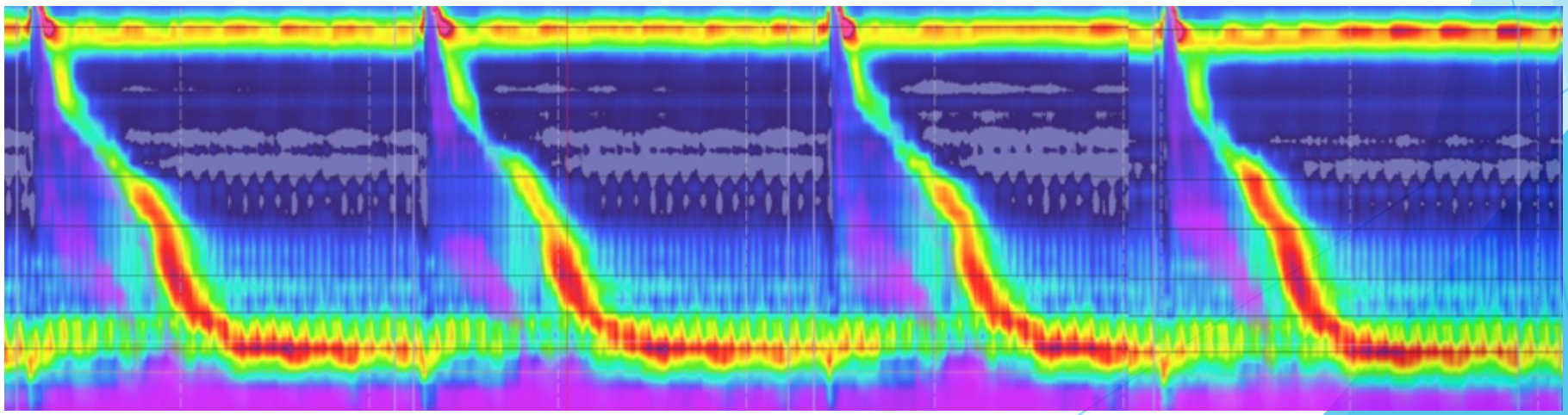
GI Motility 1

Pakkapon Rattanachaisit, MD
Department of physiology, Faculty of medicine, Chulalongkorn university

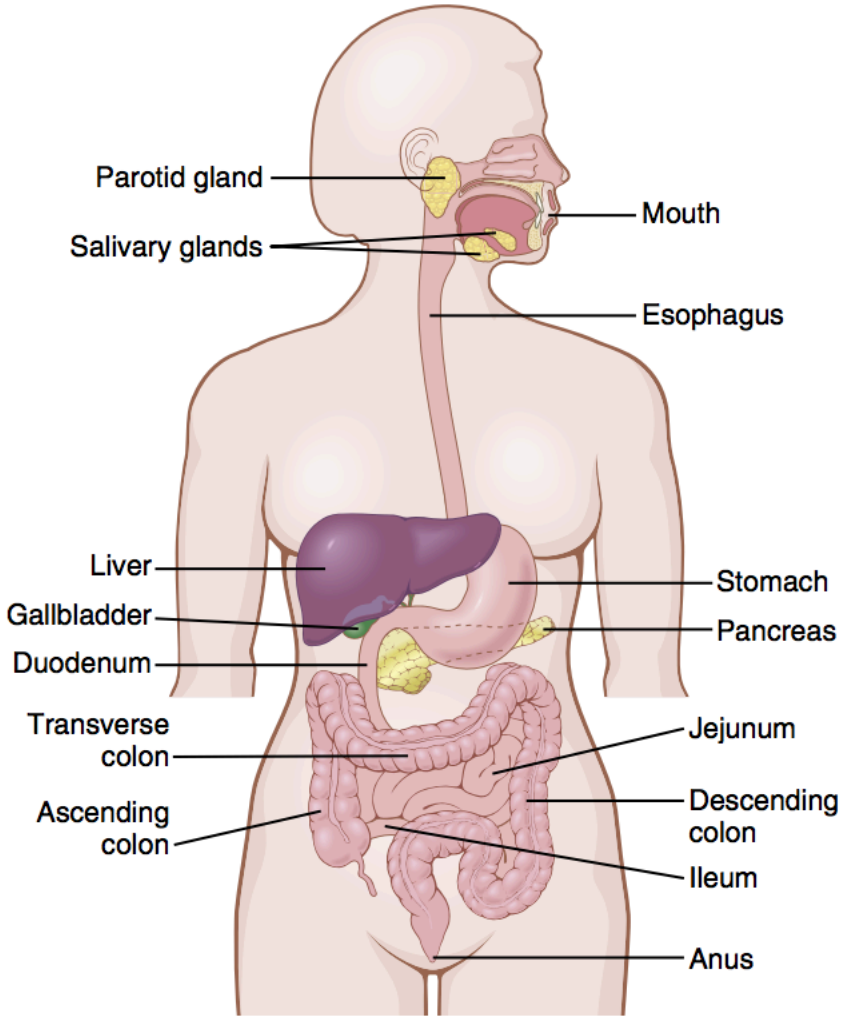


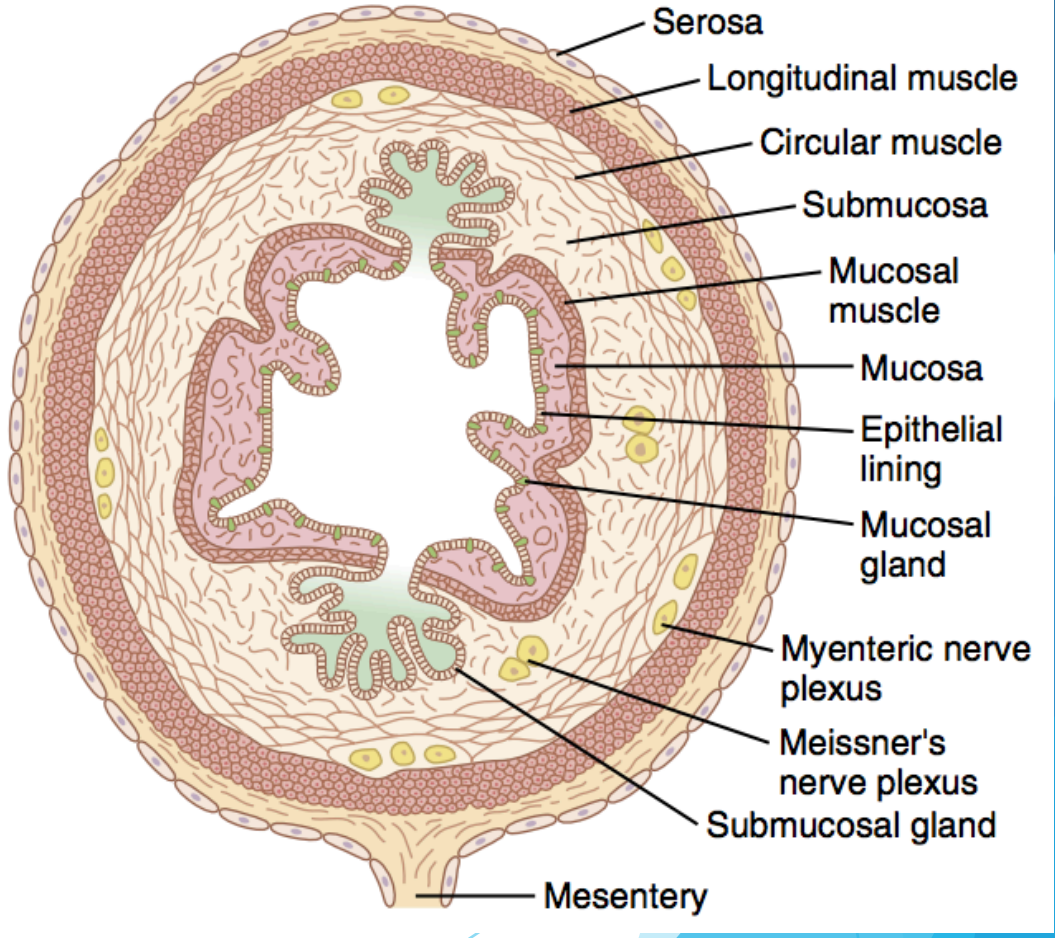
Contents

- Cellular basis and control of GI motility
- Esophageal motility
- Stomach motility
- Small intestinal motility
- Large intestinal motility
- Gallbladder and biliary tract motility

Cellular basis and control of GI motility

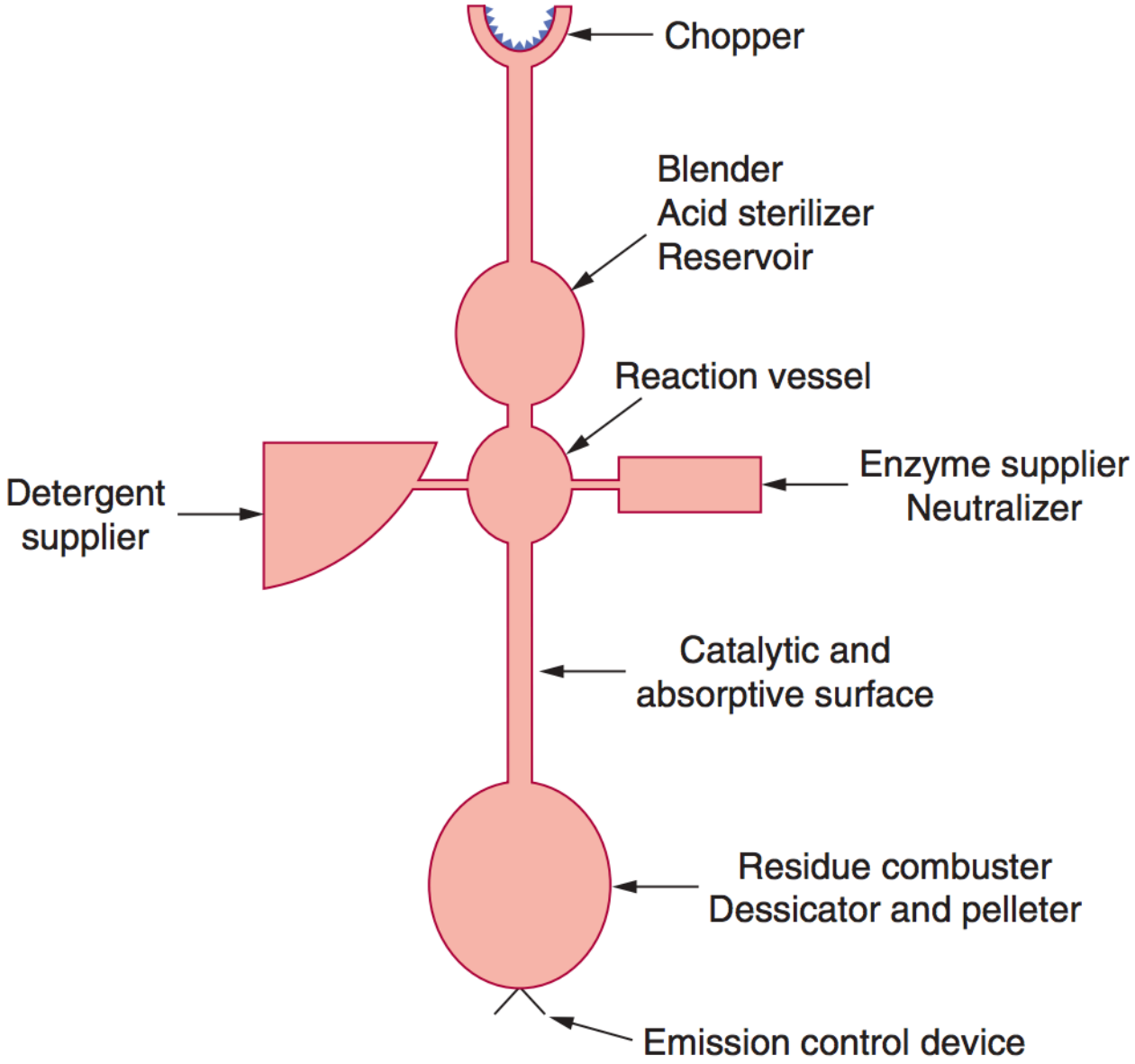
Anatomy of the GI tract





Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

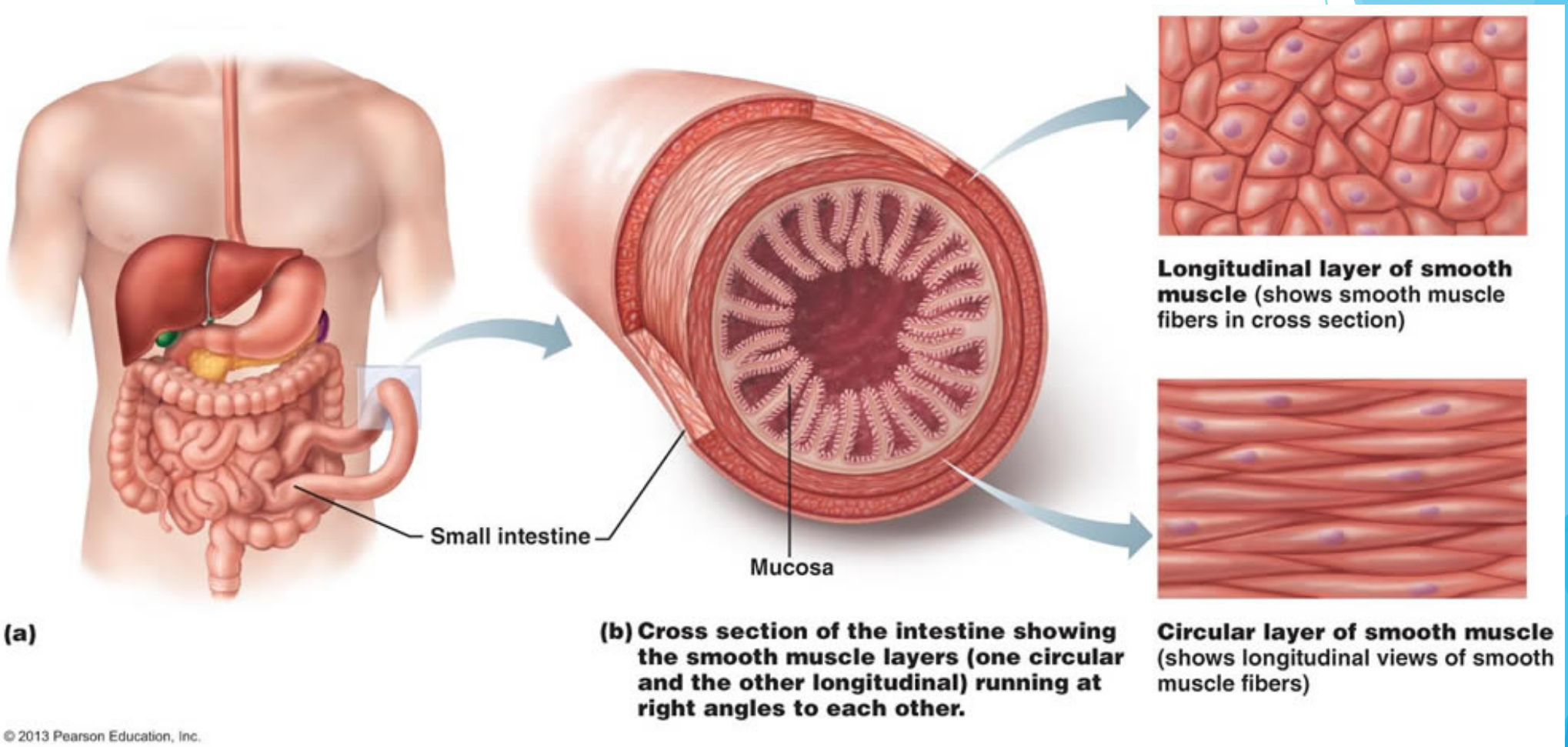
Gastrointestinal system as a machine



- GI motility
- GI secretion
- GI digestion
- GI absorption

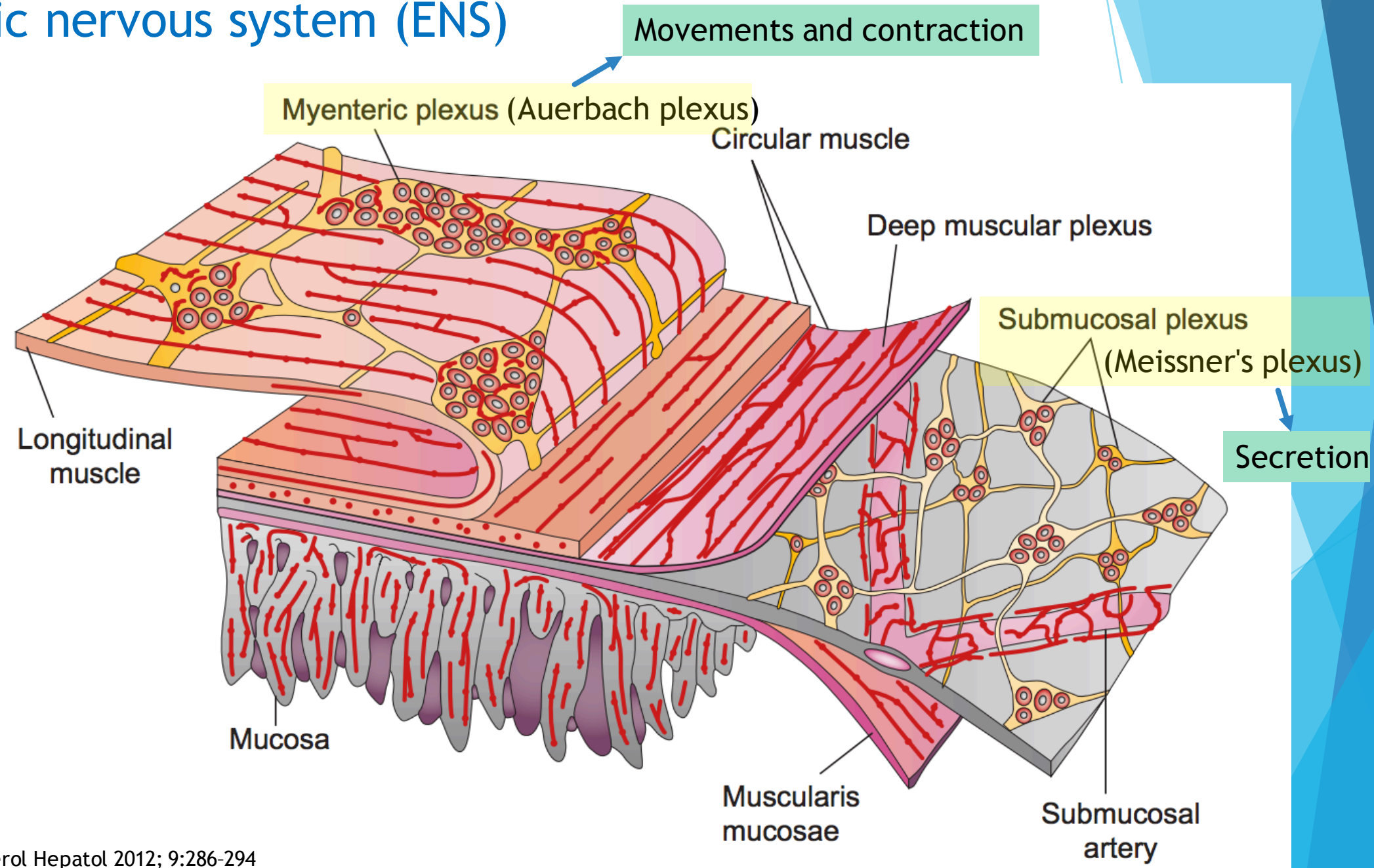
Ganong's Review of Medical Physiology, 23rd ed. McGraw-Hill Medical, 2009.

Smooth muscle of the GI tract

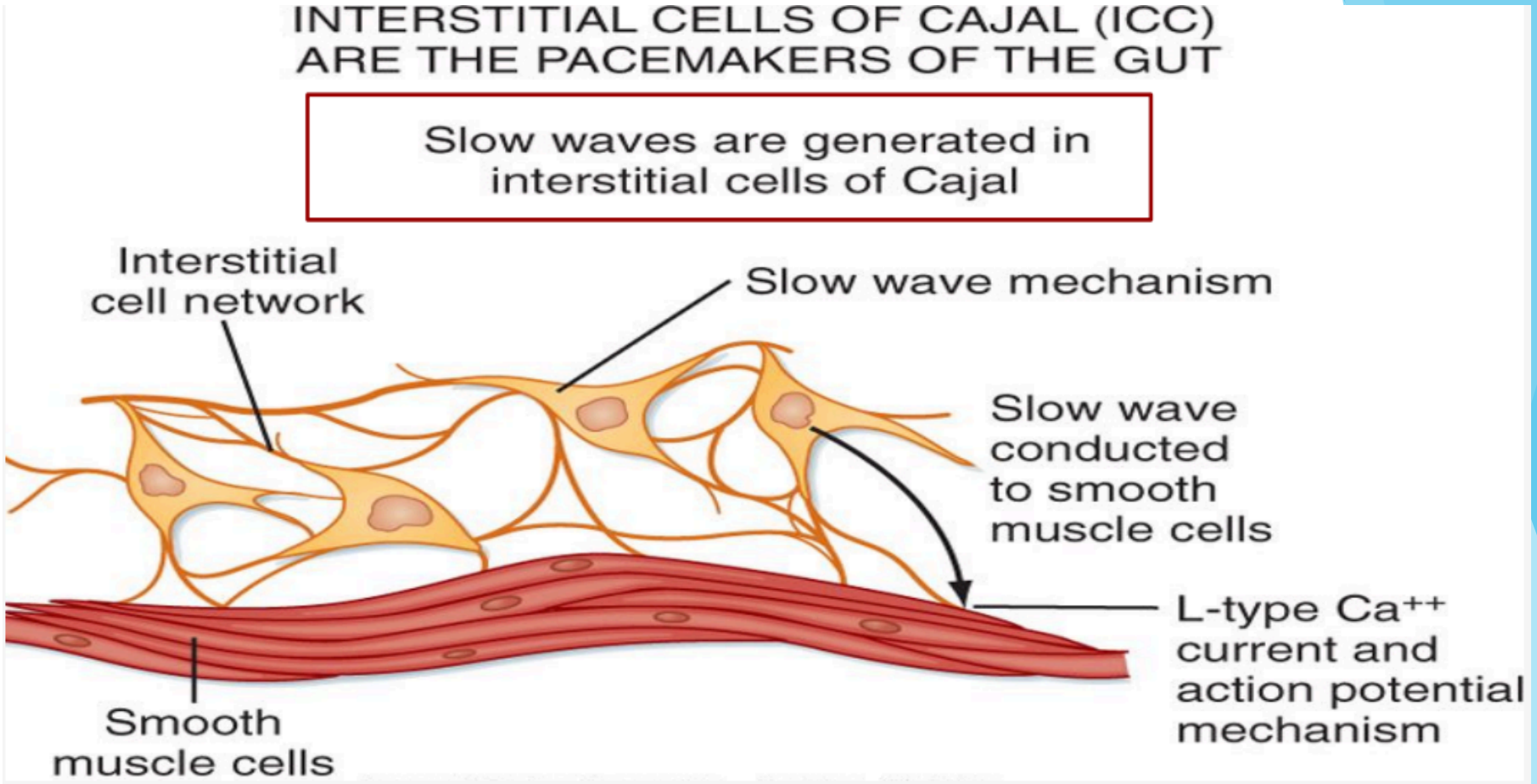


Except: pharynx, upper one-third of the esophagus, and external anal sphincter, all of which are striated muscle.

Enteric nervous system (ENS)



Nat Rev Gastroenterol Hepatol 2012; 9:286–294



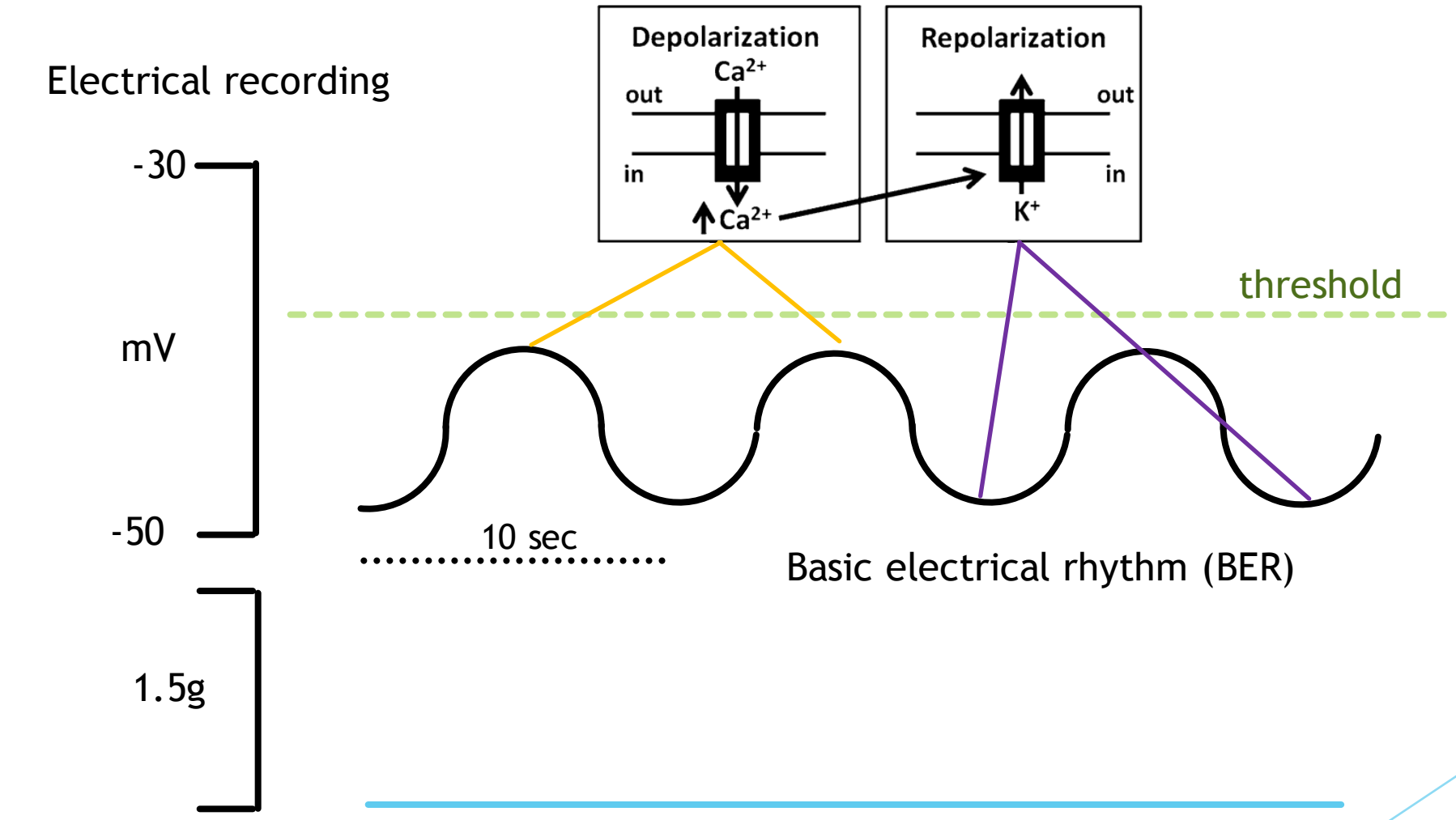
- Depolarization of circular muscle leads to contraction of a ring of smooth muscle and a decrease in diameter of that segment of the GI tract.
- Depolarization of longitudinal muscle leads to contraction in the longitudinal direction and a decrease in length of that segment of the GI tract.

Berne & Levi Physiology, th 6 Edition, 2009

Slow wave

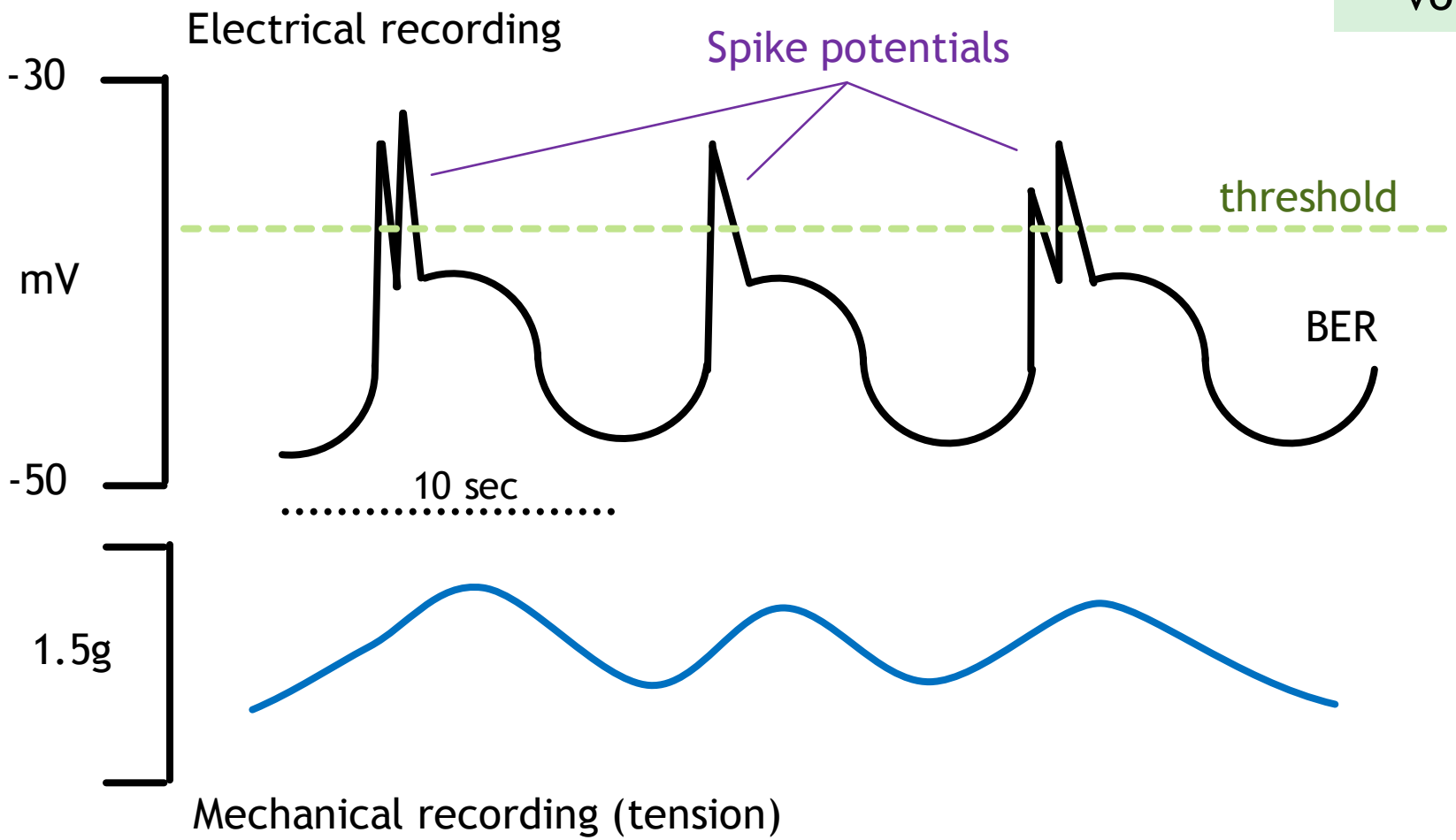
- Oscillating membrane potentials (changes in resting membrane potential)
- NOT action potentials, Do not by themselves cause muscle contractions
- Occur spontaneously from the interstitial cells of Cajal
- Determine the pattern of action potentials
- Frequency: Stomach 3/min, duodenum 12/min, ileum 8-9/min

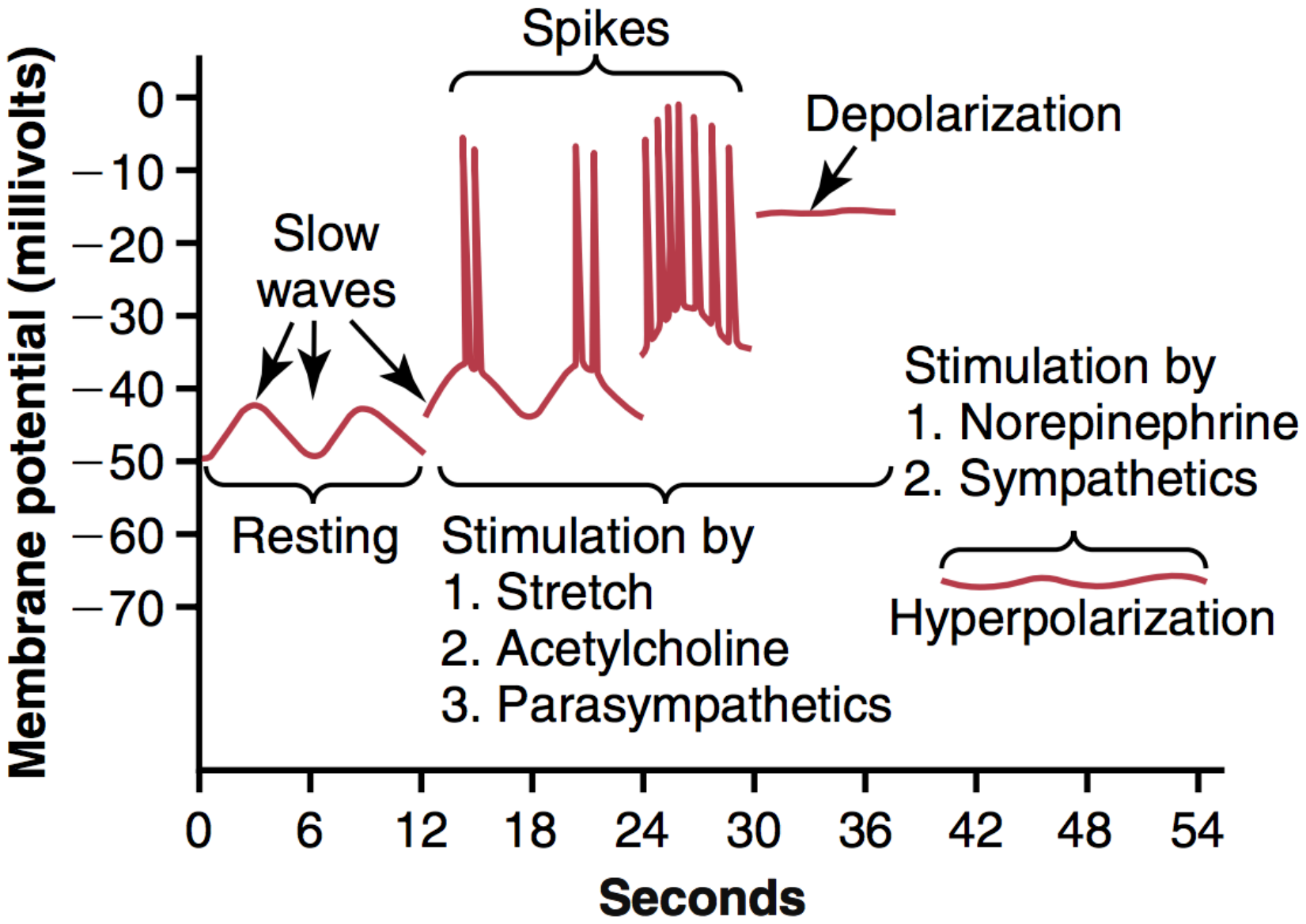
Slow wave


Depolarization
Ca2+
out
in
Ca2+
Repolarization
out
in
K+
Electrical recording
-30
mV
-50
threshold
10 sec
Basic electrical rhythm (BER)
1.5g
Mechanical recording (tension)

Spike potential

- True action potential
- Cause muscle contraction via voltage-dependent Ca^{2+} channels

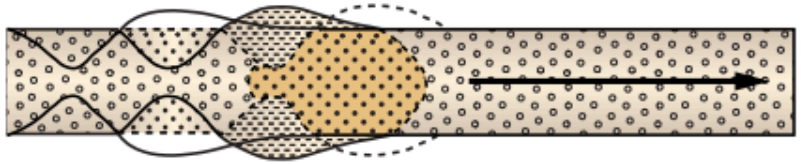
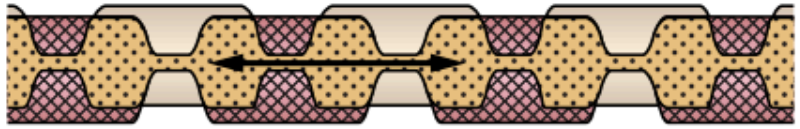
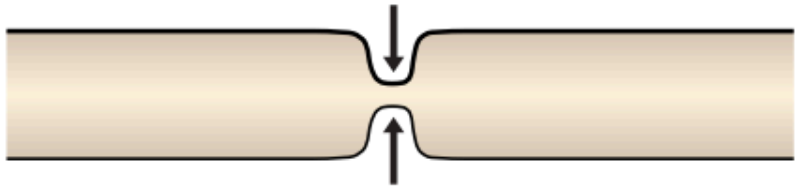




Guyton and Hall textbook of medical physiology. 14. ed., Elsevier; 2020.

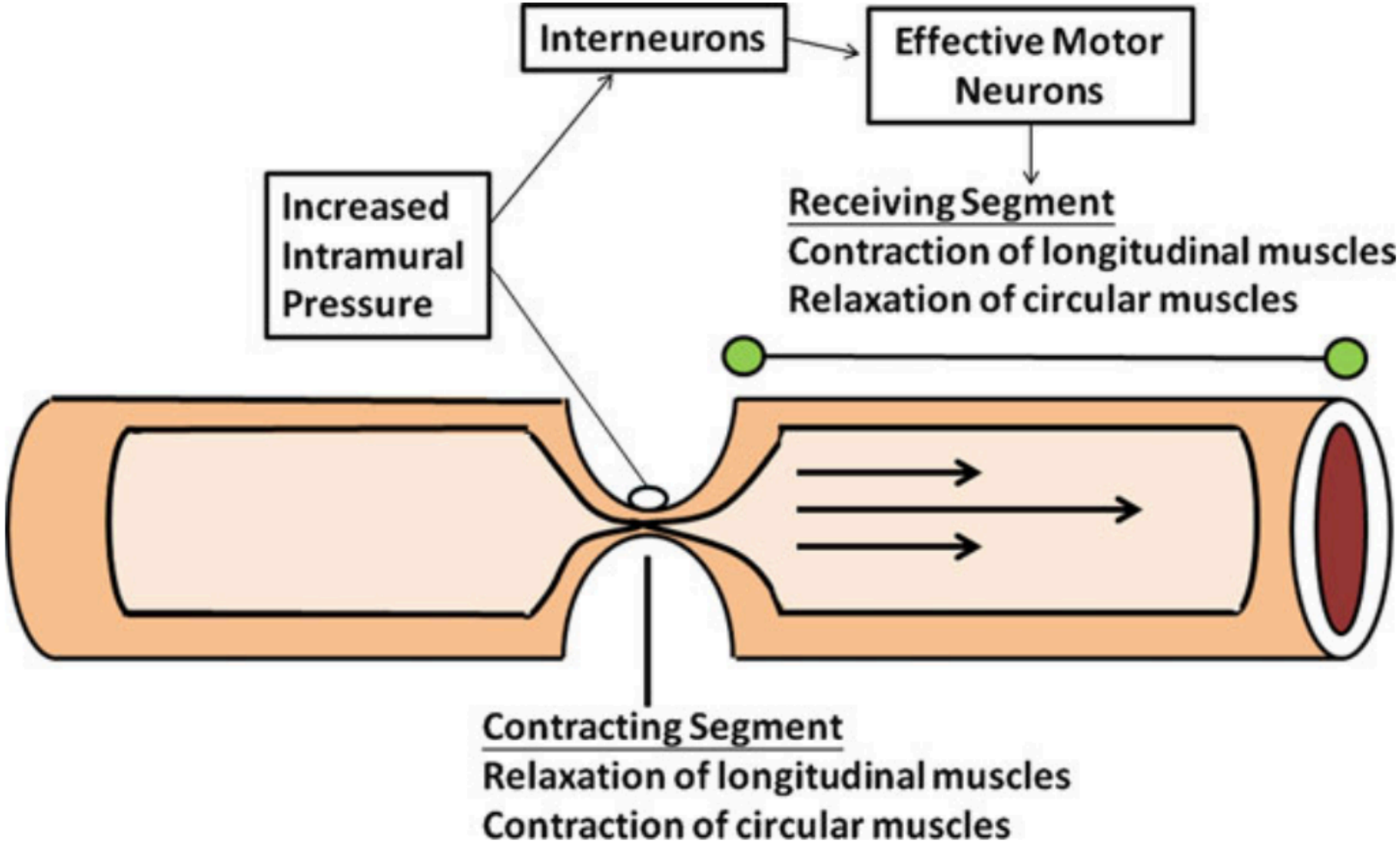
Type of contraction

Phasic contractions	Tonic contractions
• Contract and relax periodically • Occur in the esophagus, gastric antrum, and small intestine	• Maintained contraction without relaxation • Occur in the lower esophageal sphincter, orad stomach, ileocecal valve and internal anal sphincters

Motility pattern	Function	Site
 Peristalsis	Propulsive causes transport non-propulsive causes mixing	Esophagus Stomach Small intestine
 Rhythmic segmentation	Mixing	Small and large intestines
 Tonic contraction	Blocking passage Separation	Gastrointestinal sphincters

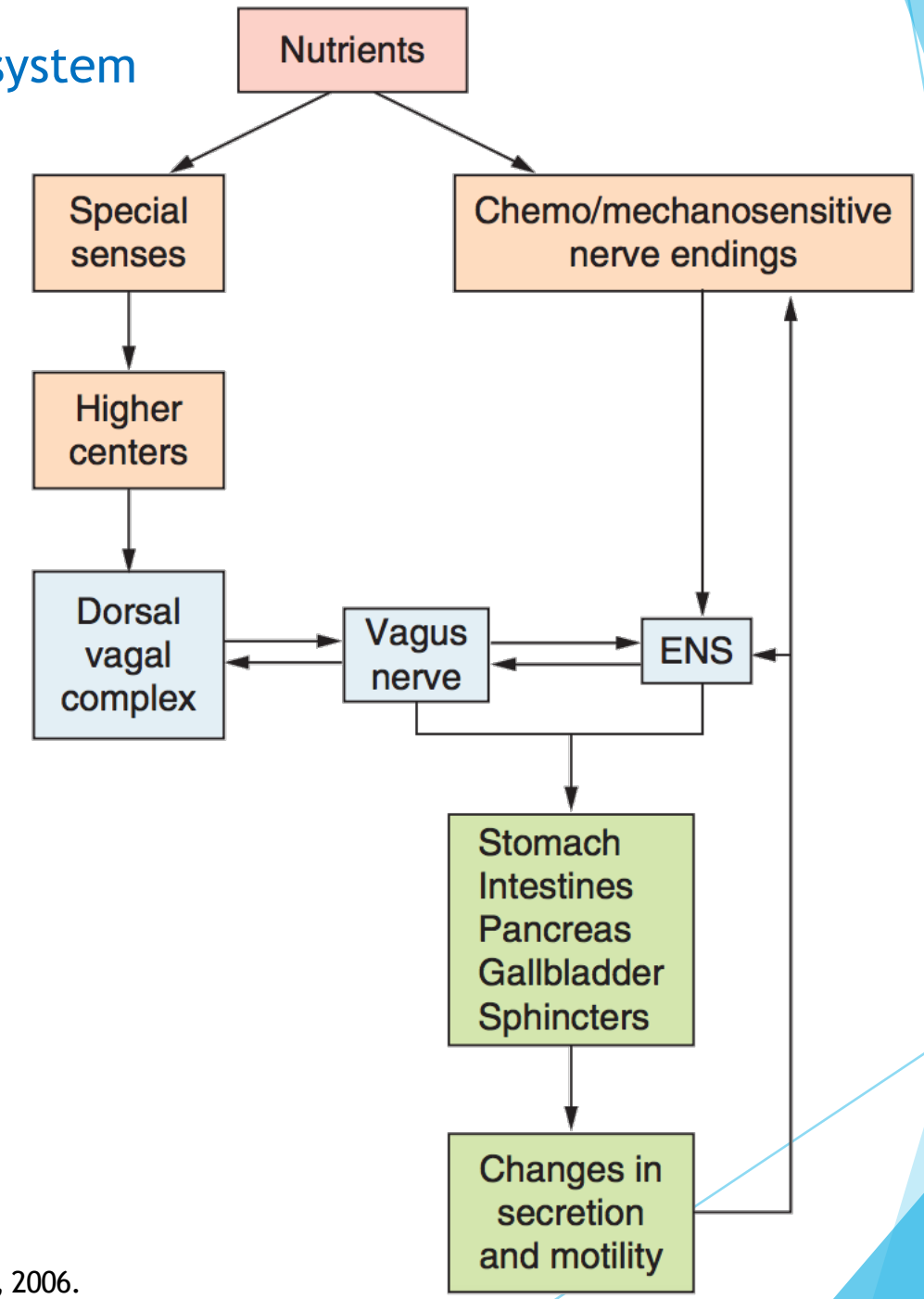
Physiology and Pathophysiology of Digestion. Morgan & Claypool; 2018.

Mechanism of peristaltic reflex



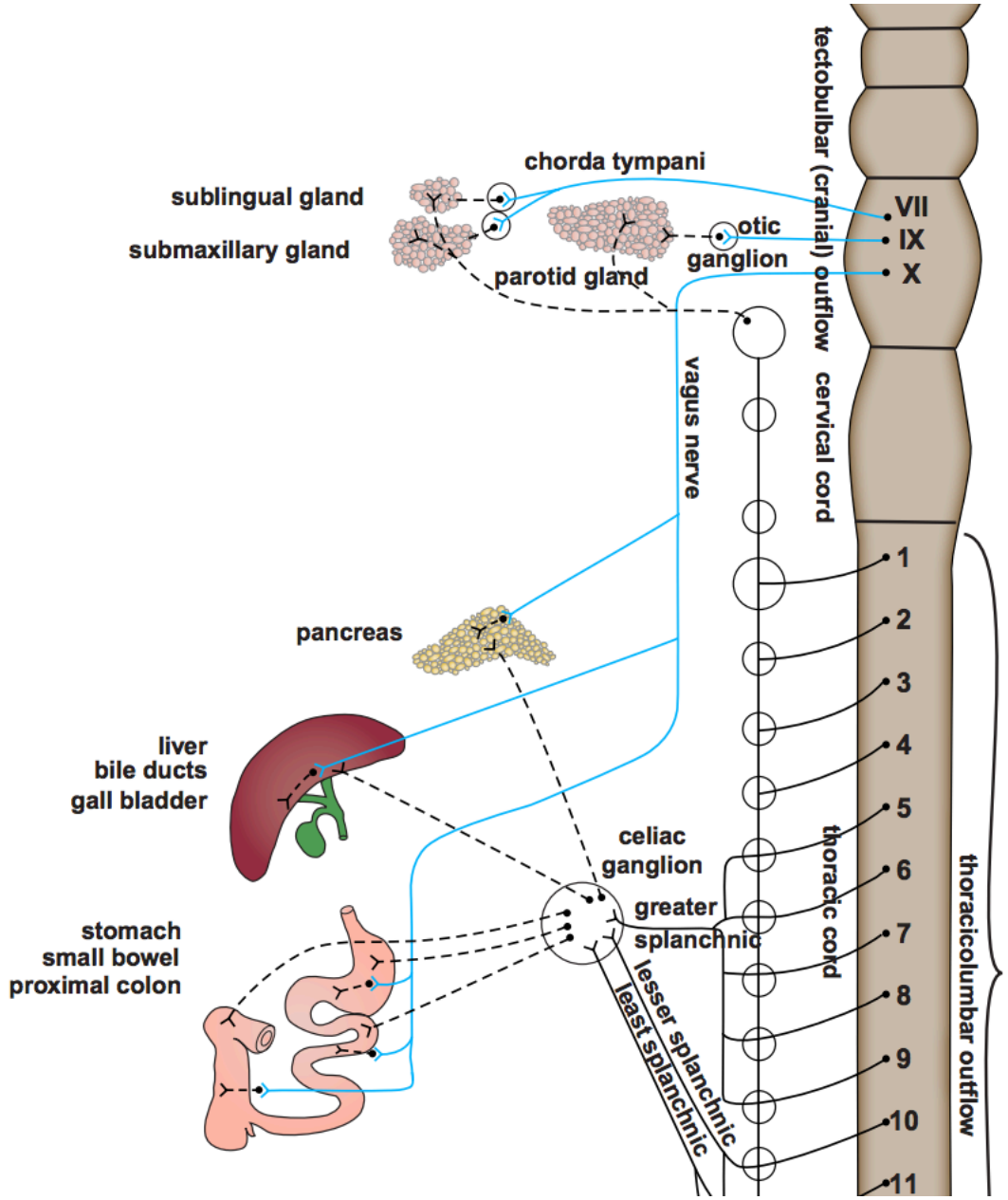
The Gastrointestinal System, Gastrointestinal, Nutritional and Hepatobiliary Physiology: Springer; 2014.

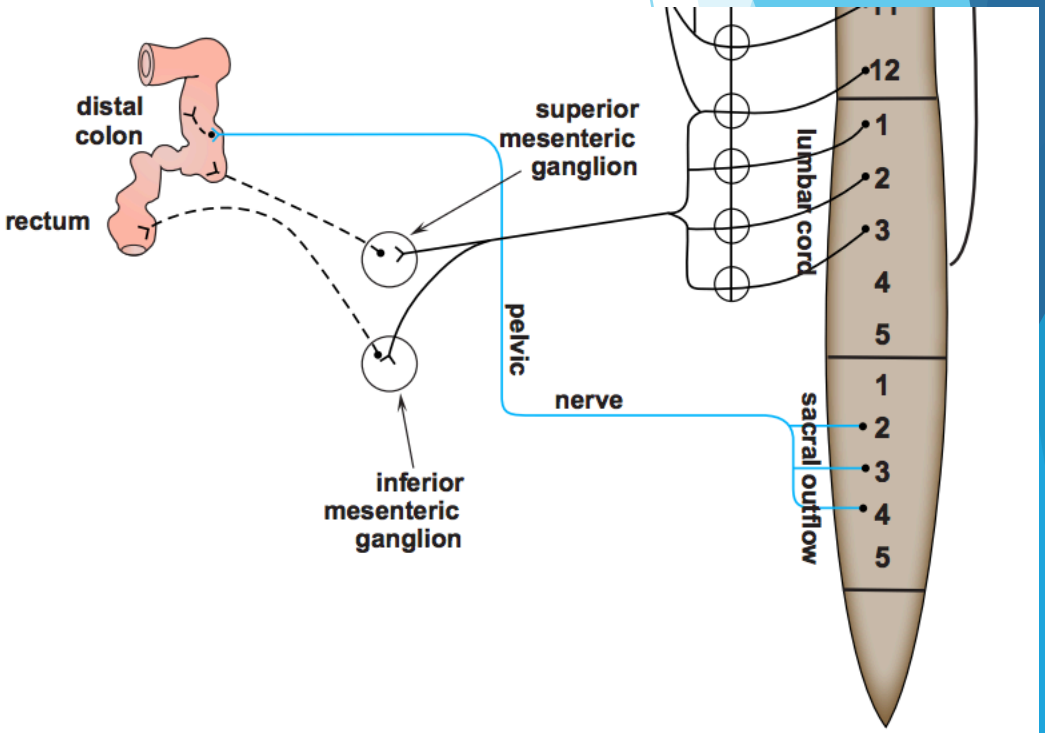
Neural control of the gastrointestinal system



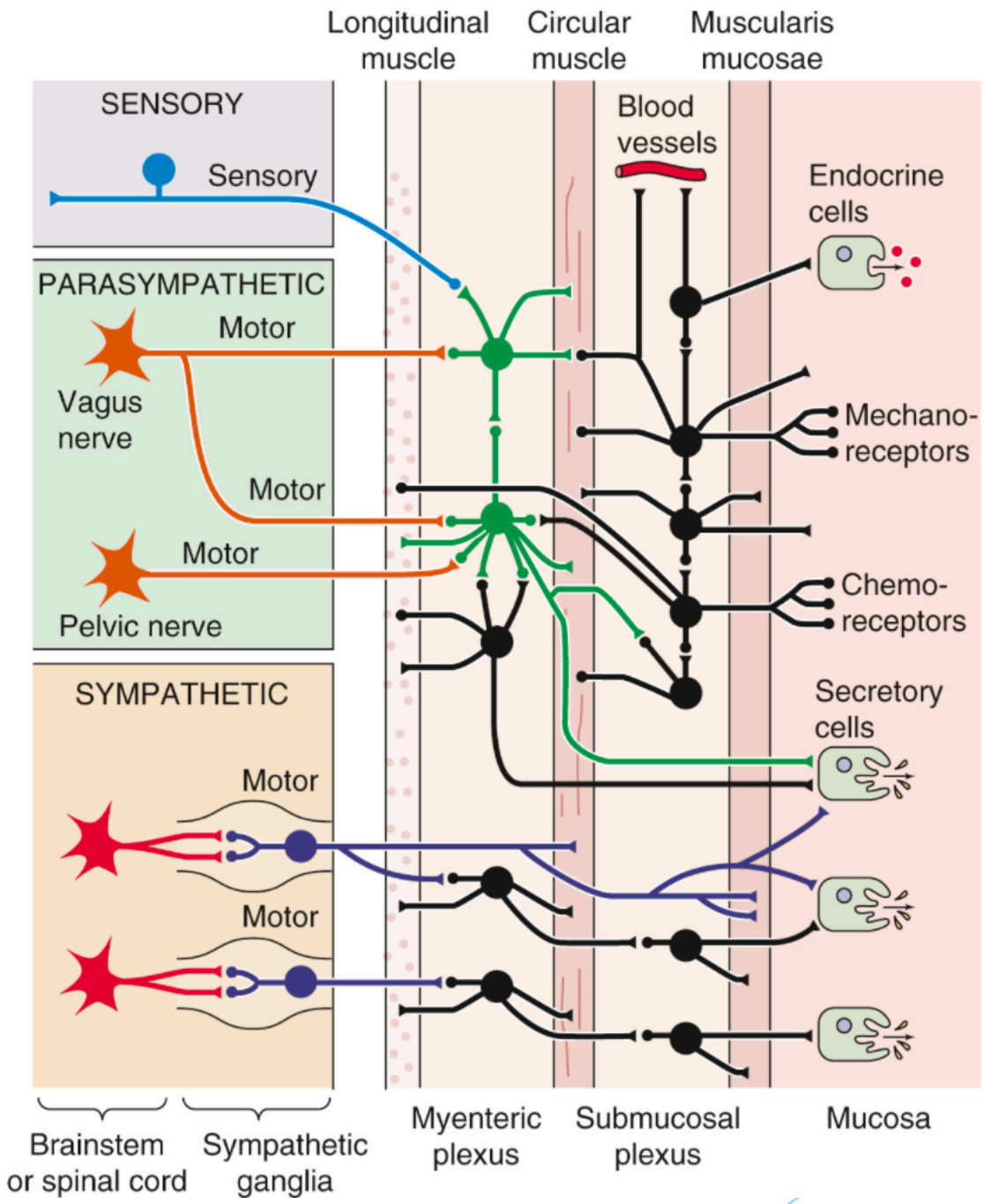
Gastrointestinal Physiology. Lange Medical Books/McGraw-Hill, 2006.

Parasympathetic and Sympathetic innervation of GI tract

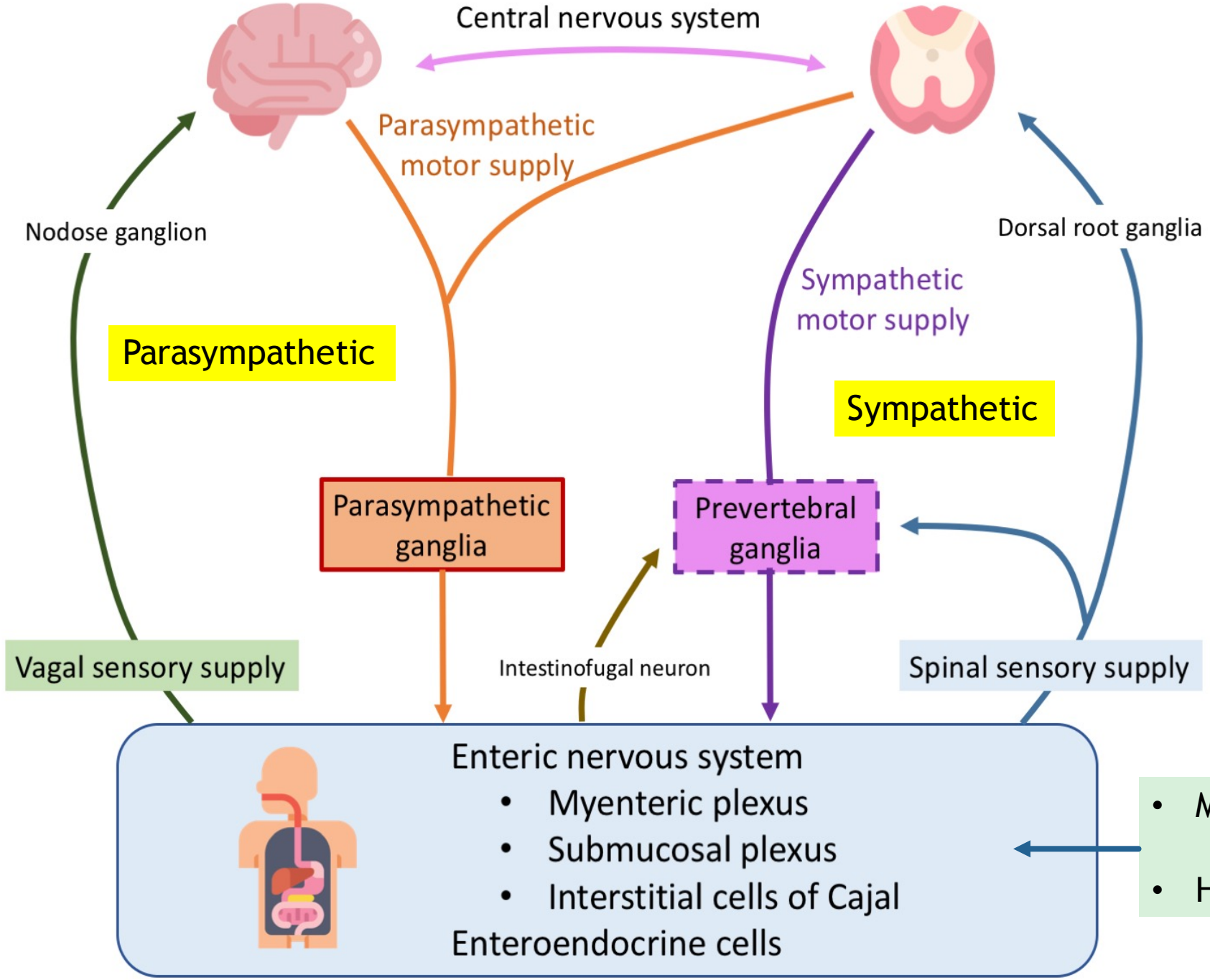




Physiology and Pathophysiology of Digestion. Morgan & Claypool; 2018.

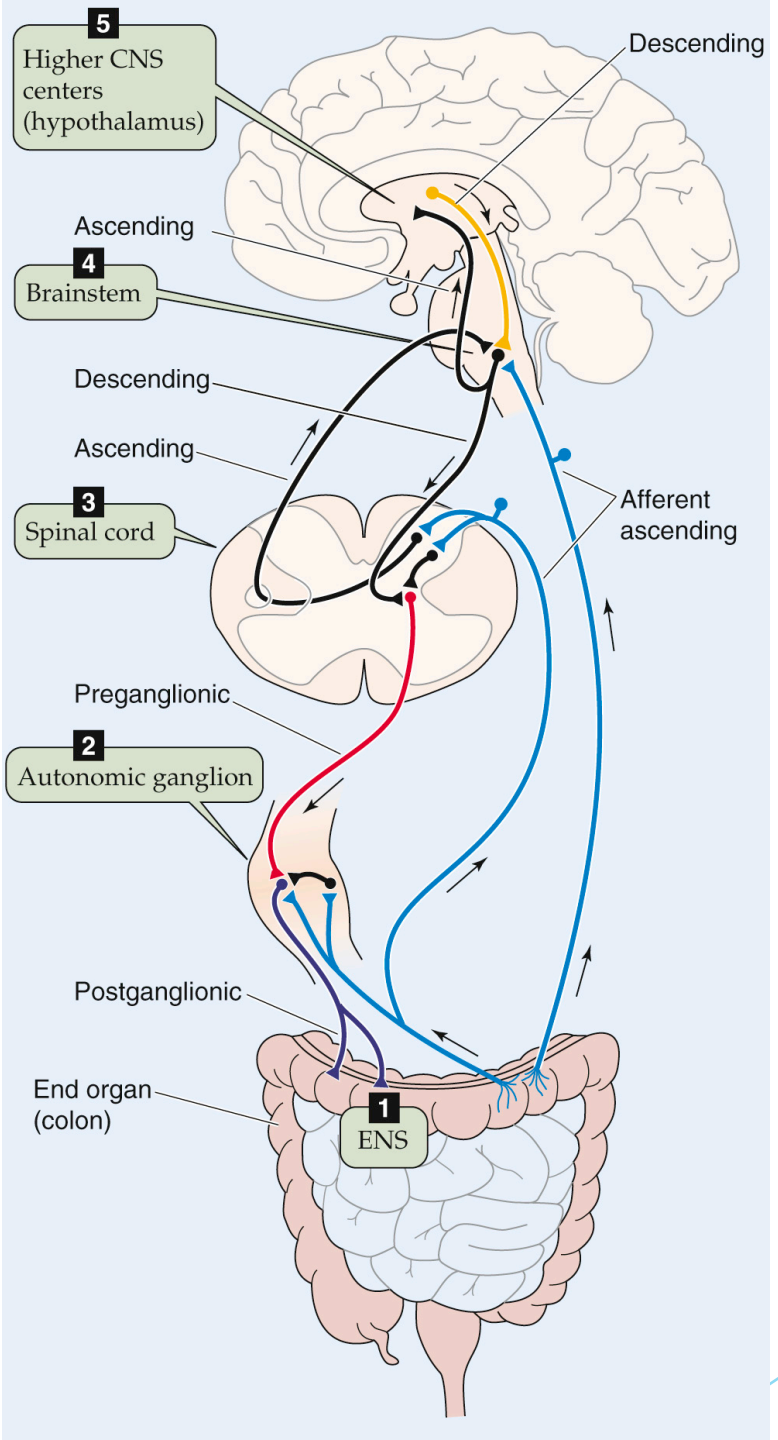


Boron W. Medical physiology, Elsevier, 2017


Central nervous system
Parasympathetic motor supply
Nodose ganglion
Dorsal root ganglia
Sympathetic motor supply
Parasympathetic
Sympathetic
Parasympathetic ganglia
Prevertebral ganglia
Vagal sensory supply
Intestinofugal neuron
Spinal sensory supply
Enteric nervous system
Myenteric plexus
Submucosal plexus
Interstitial cells of Cajal
Enteroendocrine cells
Mechanical stimulant (stretch)
Hormones

Gastrointestinal reflex

1. ENS level (GI wall): GI secretion, contraction
2. ANS level: reflex arch via prevetebral sympathetic ganglia, e.g. gastrocolic reflex, enterogastric reflex, colonoileal reflex
3. CNS level: reflex arch via brain, spinal cord, e.g. swallowing reflex, defecation reflex



Boron W. Medical physiology, Elsevier, 2017